



7th Global AI Hackathon

C H A L L E N G E

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Rental Housing Law Navigator

Which rules apply here today, and what is about to change?

POWERED BY

Realpage.

Goals and Motivation

Rental Housing Law Navigator is an AI system that reads public housing law and answers one question for any apartment address: *which rules apply here today, and what is about to change?*

THE QUESTION

How might AI turn thousands of pages of state and local housing law into accurate, cited, address-level answers that renters, advocates, housing agencies and housing providers can trust?

The problem is specific, real and getting harder

Rental housing in the U.S. is regulated in layers: state statutes, county rules and city ordinances, each with its own coverage tests (building age, unit count, owner type), effective dates and exemptions. The answer to “what rules apply to this apartment?” depends on the exact address, and it changes often.

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cities and counties across 8 states have enacted local bans on algorithmic rent-setting since late 2024, most with different definitions and penalties.

Jan 2026

California's amended antitrust law on common pricing algorithms (AB 325 / SB 763) took effect, adding a state layer on top of the local ones.

Jun 2026

Massachusetts' high court removed a statewide rent-control question from the November ballot. Proposed changes don't always become law, and tools must know the difference.

Today, getting a reliable answer means reading municipal codes, statutes and pending bills by hand. Renters don't know their rights, small housing providers don't know their obligations, and advocates and agencies can't see the full picture across jurisdictions. The source text is public, but it is unstructured, scattered and constantly changing.

Who This Helps

RENTERS

Know their rights at their own address: rent increase limits, deposit caps, fee rules, eviction protections.

ADVOCATES & AGENCIES

See which protections cover which buildings, and what a pending bill would change.

HOUSING PROVIDERS

Especially small owners without legal teams: understand their obligations before they act.

THE CHALLENGE IN ONE LINE

In 24 hours, build a working system that extracts rules from a provided corpus of real housing law, resolves any sample address to the rules that apply to it with citations, and shows which addresses a law change affects.

What Teams Build – Use the provided [data & codebase](#)

From legal text to an address-level answer

Three required modules plus stretch goals, built on a [starter pack we provide](#) so teams spend the 24 hours on AI, not data collection. Minimum entry and event rules are under **Rules, Submission and Scoring**. [Click here](#) to download the data & code.

1. Extract

Turn statutes and ordinances into structured rule records, using the provided schema.

2. Resolve

Geocode an address and build its jurisdiction stack: state, county, city.

3. Apply

Test each rule's coverage conditions against building facts: year built, units, owner type.

4. Explain

Return every applicable rule with a plain-language summary and a citation to the source text.

5. Track change

Given a new or pending law, list the addresses affected and what changes for each.

Module A: Rule extraction

Required

An agent reads each document in the corpus and outputs one record per rule, in the provided JSON format. Extraction must be automated, not hand-coded: category, jurisdiction, requirement, coverage conditions, exemptions, effective date, status (enacted or pending), penalty, source citation and quoted span.

Module B: Address lookup

Required

Given an address, return the jurisdiction stack and every applicable rule. Where a local rule overrides a state rule, the system must say so. Where coverage depends on a fact the data does not contain, it must say “unknown”, not guess.

Module C: Change tracking

Required

Run the provided change test cases. For each one, list affected sample addresses and the before/after rule set, and support an “as of date” query.

Stretch goals

- Renter-facing plain-language view in English and Spanish
- Confidence score and conflict flag for each answer
- Extend to one new jurisdiction live during the event

Illustrative output

Address lookup · as of Oct 1, 2026 — Sample: 20-unit building in San Francisco, built 1962. Jurisdictions: California › City & County of San Francisco

RENT INCREASES	SF Rent Ordinance applies (certificate of occupancy on or before 6/13/1979); the AB 1482 state cap yields to it. <i>S.F. Admin. Code ch. 37 · Cal. Civ. Code §1947.12</i>
JUST CAUSE	Eviction only for listed causes. <i>S.F. Admin. Code §37.9 · Cal. Civ. Code §1946.2</i>
DEPOSIT	Capped at one month's rent (small-landlord exception does not apply at 20 units). <i>Cal. Civ. Code §1950.5 (AB 12, eff. 7/1/2024)</i>
SCREENING FEE	Capped at a CPI-adjusted amount; no fee if no unit is available. <i>Cal. Civ. Code §1950.6</i>
ALGORITHMIC PRICING	Local ban on algorithmic devices using nonpublic competitor data, plus state restrictions on common pricing algorithms. <i>S.F. Admin. Code §37.10C (Oct 2024) · AB 325 / SB 763 (eff. 1/1/2026)</i>

Illustrative only. Teams' outputs are scored against the answer key, not this mock-up.

Scope for 24 Hours

3 states · 10 cities · 6 rule categories

Small enough to finish, varied enough to be hard. The jurisdictions were chosen to cover state-only rules, layered local rules, and recent or failed changes.

State	Cities in scope	Why it's in the set
California	Los Angeles, San Francisco, San Diego, Berkeley, Santa Ana*	Densest layering: statewide rent cap and just cause, local rent control, statewide and local algorithmic-pricing rules with different effective dates.
New Jersey	Jersey City, Hoboken, Newark	Rent control is set city by city; statewide eviction, screening and fee rules; two local algorithmic-pricing bans and a new statewide FAIR Act (effective July 2027) that may preempt them.
Massachusetts	Boston, Cambridge	State law bars local rent control; a 2026 ballot question was struck before the vote; algorithmic-pricing bills are pending. Tests whether systems avoid reporting rules that don't exist.

**Santa Ana laws are in the corpus for extraction, but no open parcel data with addresses exists, so the address sample covers the other 9 cities.*

Rule categories, with real examples from the corpus

Category	What to capture	Real examples teams will encounter
1. Rent increase limits	Cap formula, covered buildings, exemptions, local vs. state precedence	CA Tenant Protection Act, Civ. Code §1947.12 (5% + CPI, max 10%) · SF Rent Ordinance, Admin. Code ch. 37 · LA Rent Stabilization Ordinance · MA G.L. c.40P (state bar on local rent control)
2. Just-cause eviction	Allowed causes, notice, relocation assistance, coverage	CA Civ. Code §1946.2 · NJ Anti-Eviction Act, N.J.S.A. 2A:18-61.1
3. Security deposits	Maximum, exceptions, effective date	CA Civ. Code §1950.5 as amended by AB 12 (one month; two for qualifying small landlords; eff. 7/1/2024) · NJ N.J.S.A. 46:8-21.2 (1.5 months) · MA G.L. c.186 §15B (first month's rent)
4. Application & screening fees	Fee caps, allowed upfront charges, receipts and refunds	CA Civ. Code §1950.6 (CPI-adjusted cap) · NJ P.L.2025, c.405 (\$50 cap, eff. 5/1/2026) · MA G.L. c.186 §15B (upfront charges limited to first and last month's rent, deposit, lock) · MA broker-fee rule, G.L. c.112 §87DDD½ (8/1/2025)
5. Screening restrictions	Limits on criminal-history and income-source screening; timing rules	NJ Fair Chance in Housing Act (2021) · CA source-of-income protections under FEHA (SB 329)
6. Algorithmic rent-setting	Definition of covered software, prohibited conduct, penalties, effective date	CA AB 325 / SB 763 (1/1/2026) · San Francisco §37.10C (Oct 2024) · San Diego §§98.1101–98.1104 (Jun 2025) · Berkeley ch. 13.63 (2026) · Santa Ana Ord. NS-3090 (Apr 2026) · Jersey City §218-12 (Jun 2025) · Hoboken ch. 158, Art. II (Jul 2025) · NJ FAIR Act, P.L.2026, c.43 (eff. 7/1/2027) · MA S.2983 / H.5222 (pending)

The answer key holds 58 rules and 19 “no rule at this level” findings; 52 of the 58 rules were verified against public sources as of October 1, 2026. It has not been reviewed by counsel. This brief summarizes these laws for scoping only and is not legal advice.

Test Cases and Starter Pack

Real law changes, packaged data

Change tracking is tested on real 2025–2027 changes, real pending bills and one fictional ordinance revealed mid-event. To make 24 hours realistic, Realpage pre-packages a starter kit from public sources, so teams don't lose the first day to scraping and cleanup. [Everything is in one Google Drive folder which you can access here.](#)

Change-tracking test cases

Test case	What a correct system does
T1 · CA AB 325 / SB 763, effective 1/1/2026	"Not yet effective" for CA addresses as of 12/31/2025; "applies" as of 1/2/2026
T2 · Hoboken and Jersey City local bans	Each ban only inside its own city limits; neither in Newark
T3 · NJ FAIR Act, signed 7/20/2026, effective 7/1/2027	"Not yet effective" today, "applies" on 7/2/2027; flags a possible conflict with the two local bans
T4 · MA S.2983 and H.5222 (pending bills)	Reports them as pending, never in force; lists the addresses they would affect
T5 · MA rent-control ballot question, struck 6/23/2026	Reports no rent cap for Boston or Cambridge; affected set is empty
T6 · Fictional Cambridge ordinance (released at hour 16)	Extracts it unaided, lists affected addresses, and gets its future effective date right

Starter pack provided to teams

Item	Contents	Format
Law corpus	87 source documents: official statute, ordinance and bill text as plain text with URL and retrieval date; law-firm and news pages as links only	Text + manifest CSV
Sample addresses	~500 multifamily properties in 9 cities from public assessor data: street, postal city, ZIP, year built, units, use code. No owner names. Teams resolve the legal jurisdiction themselves.	CSV
Rule schema	Required fields for every rule record, the 6 categories, and a worked sample record	JSON Schema
Dev answer key	10 rules and expected results for 20 addresses, for self-testing	JSON
Held-out answer key	The full key (58 rules, 19 "no rule" findings) and expected results for 100 of the 500 addresses	Held by judges
Change test cases	The 6 tests above; expected affected addresses held by judges	JSON
Scoring script	The exact script judges use: extraction, address coverage, citations, change tracking (see Scoring)	Python

Data

Everything is public

Every source below is publicly available and free. Teams can go beyond the starter pack using these. No customer, pricing or proprietary data is used.

Public sources teams may use directly

Source	What it provides	Access & terms
State codes	Official statute text: CA Legislative Information · NJ Legislature · MA General Laws (malegislature.gov)	Free, public websites
Municipal codes	Ordinance text for each city: city sites; hosted by publishers such as American Legal, Municode, eCode360	Free to read; bulk scraping often restricted by site terms. Use the starter corpus.
LegiScan API	Bills, status, full text, votes for all 50 states	Free key; 10,000 queries/month; CC BY 4.0; weekly bulk datasets
Open States API v3 (Plural)	Bills, sponsors, events by state	Free key
Census Geocoder	Address → coordinates, state, county, county subdivision, incorporated place	No key; batch up to 10,000 addresses
Census TIGER/Line	City and county boundary shapefiles	Free download
Parcel data	Building facts: year built, units, use code. MassGIS Property Tax Parcels · NJ Parcels & MOD-IV Composite · LA County and SF assessor open data	Free public data. Gaps: no year built for San Diego; no year built or units for Berkeley; no open data for Santa Ana
Local program lookups	Spot-check rent-control coverage for a property: LA Housing Dept. RSO lookup · SF Rent Board	Public websites, for validation only

LSC Eviction Laws Database	Coded eviction laws, all 50 states plus 30 local jurisdictions	Free Excel download. Current only as of 1/1/2021: use for methods, not as current truth.
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NOT PROVIDED, NOT ALLOWED	No customer or resident data, no rent or pricing data, no internal legal analysis, and no scraping that violates a site's terms of use.
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Responsible AI by Design

Build for transparency, not legal verdicts

The system should make the law easier to see and understand, without pretending to be a lawyer or helping anyone work around the rules.

THE SOLUTION SHOULD - Cite the source text and retrieval date for every rule it reports. - Show an “as of” date on every answer and separate enacted from pending law. - Say “unknown” when coverage depends on facts it doesn’t have. - Flag conflicts and low-confidence answers for human review. - Explain rules in plain language that a renter can act on. - Keep an auditable log of sources, model outputs and changes.	THE SOLUTION MUST NOT - Present output as legal advice or a compliance certification. - Suggest ways to avoid, structure around or evade a rule. - Invent rules or citations where the source text is silent. - Use customer, resident, pricing or other non-public data. - Scrape sites in violation of their terms of use.
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24-hour plan – after 16 h we will release an ad

Hours	Activity	Hours	Activity
Hr 0–1	Kickoff, starter-pack walkthrough, mentor intros	Hr 16–20	Change tracking; synthetic ordinance test data released at hour 16 here, we like you to react to it!
Hr 1–6	Rule extraction against the corpus; self-check on the dev key	Hr 20–23	Scoring run, fixes, demo prep
Hr 6–11	Geocoding, jurisdiction stacks, coverage logic	Hr 23–24	Demos and judging
Hr 11–16	Address lookup with citations and plain-language view		

From Hackathon to Impact

The strongest teams may be considered for follow-on prototyping, talent conversations, and a potential open public-good release of the rule dataset with Realpage and housing partners.

THE INTENDED OUTCOME	Show that responsible AI can make housing law visible at the level that matters, a single address, so renters know their rights, providers know their obligations, and everyone can see what is changing, with every answer traceable to the source.
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Rules, Submission and Scoring

How the event works

The participant guide in the [starter pack](#) has the full detail. These are the rules every team needs on day one.

RULES OF THE EVENT Minimum viable entry: Modules A and B, scored on the dev set. Automated extraction only. The hour-16 ordinance and a live rerun in the demo check this. “Unknown” is a valid answer when coverage depends on facts not in the data, and it earns credit. No non-public data; no scraping against a site’s terms. Every interface says “not legal advice.”	WHAT TEAMS SUBMIT rules.json: rule records in the provided format, with citation and quoted source text lookups.json: for all 500 addresses, each rule’s result: applies, unknown, superseded, not yet effective, or pending changes.json: affected addresses (and conflict flags) for each test Three short videos which need to include your scores
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Additional submission details

- **Team video:** introduce your team.
- **Demo video:** show your tool in use.
- **Technical video:** walk through how your system works.
- **GitHub repository:** your code, a README explaining how to run it, and your output files (rules.json, lookups.json, changes.json).
- **Live demo link:** a working link to your tool.

Report your test score results in your videos

- Run score.py on the dev set and show the full score report on screen.
- Show your results for change tests T1–T6.
- Show your system processing the hour-16 ordinance test dataset – you will receive this 16 h into the hackathon via the [Google Drive folder](#)

Scoring

Component	Points	How it's measured
Extraction accuracy	25 · auto	Team rules matched to the held-out key by jurisdiction, category and citation; field accuracy on date, status, key value, citation
Address coverage	20 · auto	Results on 100 held-out addresses; missing a rule that applies costs twice as much as other errors; "unknown" earns partial credit
Citations	15 · auto	Share of "applies" answers backed by a source and a quoted span found in the corpus
Change tracking	15 · auto	Overlap with the expected affected-address sets for T1–T6, plus conflict flags on T3
Plain language and usability	10 · judges	Demo
Responsible design	10 · judges	Uncertainty, audit trail, guardrails
Scalability path	5 · judges	How the approach extends to new jurisdictions

Teams self-test with the same script against the dev key. Full matching and partial-credit rules are in the participant guide.